

SAT 11

Math Module 3

Question #

ID

SAT11 M 3.1

a11ma301

Josie purchased a pass that cost \$19.50 to tour a nature preserve. She also purchased 10 identical stickers to give to her nieces and nephews. The pass and the stickers cost \$62.00 total. What was the cost of one sticker?

- A. \$1.95
- B. \$4.25
- C. \$6.20
- D. \$8.15

SAT11 M 3.2

a11ma302

$$x + y = 125$$

$$x + y + y = 155$$

The solution to the given system of equations is (x, y) . What is the value of y ?

SAT11 M 3.3

a11ma303

Leo goes to a packing store to buy containers and tape. Leo has \$15. Each container costs \$1.87 and each roll of tape costs \$2.40. Which inequality represents the relationship between the number of containers, c , and the number of rolls of tape, t , Leo can buy?

- A. $1.87c + 2.40t \leq 15$
- B. $1.87c + 2.40t \geq 15$
- C. $2.40c + 1.87t \leq 15$
- D. $2.40c + 1.87t \geq 15$

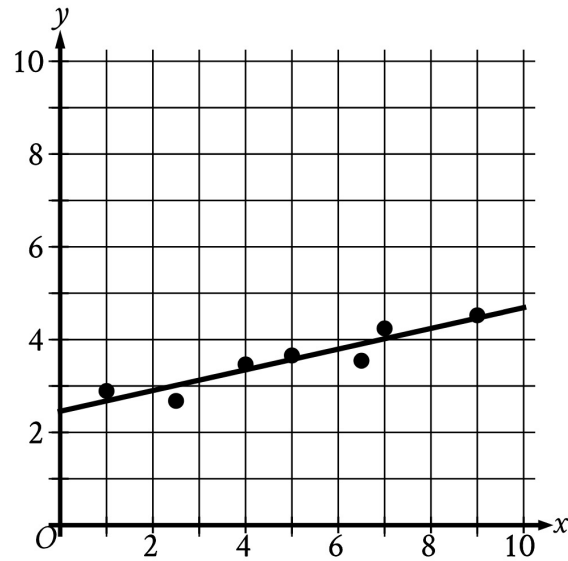
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Question # ID

SAT11 M 3.4 allma304

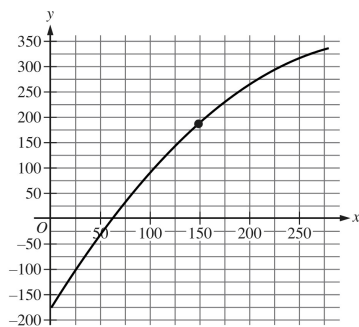
In the given scatterplot, a line of best fit for the data is shown.



- A. -2.45
- B. -0.22
- C. 0.22
- D. 2.45

Which of the following is closest to the slope of the line of best fit?

SAT11 M 3.5 allma305



The graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(149.02, 186.05)$?

- A. A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.
- B. A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.
- C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.
- D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

SAT11 M 3.6 allma306

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

- A. -4
- B. 8
- C. 42
- D. 44

Question # ID
SAT11 M 3.7 allma307

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

SAT11 M 3.8 allma308

The function f is defined by $f(x) = 9(2x + 3)$. For what value of x does $f(x) = 63$?

- A. 2
- B. 5
- C. 7
- D. 30

SAT11 M 3.9 allma309

$$x^2 + 7x + 5 = 0$$

One solution to the given equation can be written as $x = \frac{-7 + \sqrt{k}}{2}$, where k is a constant. What is the value of k ?

SAT11 M 3.10 allma310

$$(9x^4 + 8x^3) - (7x^3 + 5x^2) - (3x^2 + x)$$

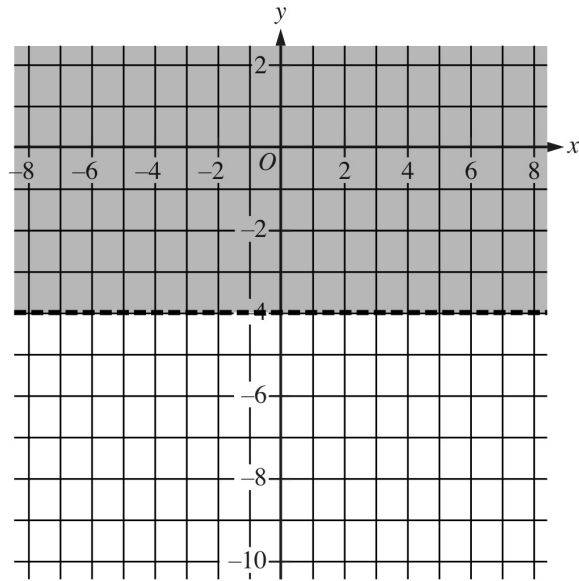
The given expression can be written in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. What is the value of a ?

- A. -8
- B. -2
- C. 2
- D. 8

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Question # ID
SAT11 M 3.11 allma311



The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r ?

- A. 15
- B. 4
- C. -4
- D. -15

SAT11 M 3.12 allma312

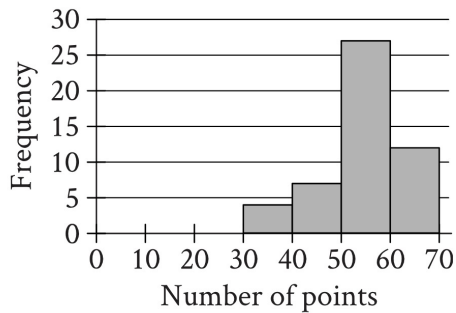
Set K consists of all the positive integers that are less than or equal to 160, where none of the integers are repeated. If a number is selected at random from set K, what is the probability of selecting a number that is even and less than or equal to 50?

- A. $\frac{5}{32}$
- B. $\frac{3}{16}$
- C. $\frac{5}{21}$
- D. $\frac{5}{16}$

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Question # ID
SAT11 M 3.13 a11ma313



The histogram summarizes data set A, which represents the number of points per player earned by 50 players of a game. A new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. Which of the following must be true?

- I. The median number of points per player for data set B is less than the median number of points per player for data set A.
- II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

SAT11 M 3.14 a11ma314

In a scale model of an office building, a length of 1 inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. If a length in the actual office building measures x feet, which expression represents the corresponding length, in inches, in the model in terms of x ?

- A. $\frac{3x}{(17)(12)}$
- B. $\frac{17x}{(12)(3)}$
- C. $\frac{3x}{17}$
- D. $\frac{17x}{3}$

SAT11 M 3.15 a11ma315

$$\sqrt[3]{p^2} = t^{\frac{9}{7}}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{3n-1}$, where n is a constant, what is the value of n ?

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Question # ID
SAT11 M 3.16 a11ma316

$$g(x) = \frac{x^2 - x - a}{x^3 - x - b}$$

The function g is defined by the given equation, where a and b are constants. In the xy -plane, the graph of $y = g(x)$ passes through the point $(0, 22)$, and $g(-22) = 0$. What is the value of b ?

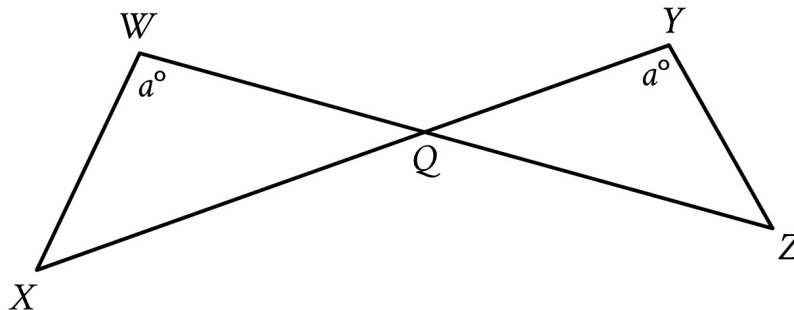
- A. 23
- B. 22
- C. -22
- D. -23

SAT11 M 3.17 a11ma317

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 140)^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

- A. $(-18x + 280)^\circ$
- B. $(-18x + 640)^\circ$
- C. $(18x - 280)^\circ$
- D. 180°

SAT11 M 3.18 a11ma318



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

- A. 18
- B. 36
- C. 120
- D. 200

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Question # ID
SAT11 M 3.19 a11ma319

$$2x + 9y = 7$$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

- I. $3x + 13.5y = 10.5$
- II. $3x - 13.5y = 10.5$

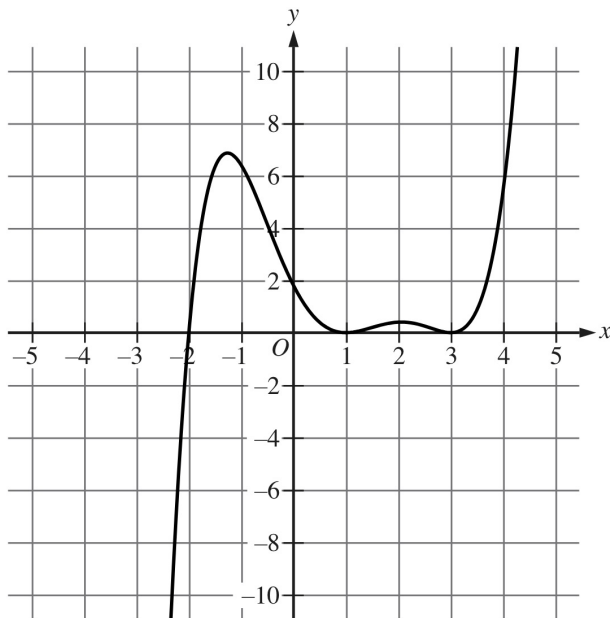
- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

SAT11 M 3.20 a11ma320

A right rectangular prism has a base area of $24t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the rectangular prism is 15 cm. Which expression represents the surface area, in cm^2 , of the right rectangular prism?

- A. $48t + 160$
- B. $318t + 80$
- C. $1,968t + 80$
- D. $360t$

SAT11 M 3.21 a11ma321



The graph of $y = g(x)$ is shown. If $g(x + 28) = \frac{1}{10}(x - a)^2(x - b)^2(x - c)$, where a , b , and c are constants, which of the following could be the value of b ?

- A. 31
- B. 3
- C. -2
- D. -25

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Question #

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SAT11 M 3.22

a11ma322

The weight of sample A is 468% of the weight of sample B, and the weight of sample A is 0.780% of the weight of sample C. If the weight of sample C is $p\%$ of the weight of sample B, what is the value of p ?